

L1.5 Differentiation Formulae

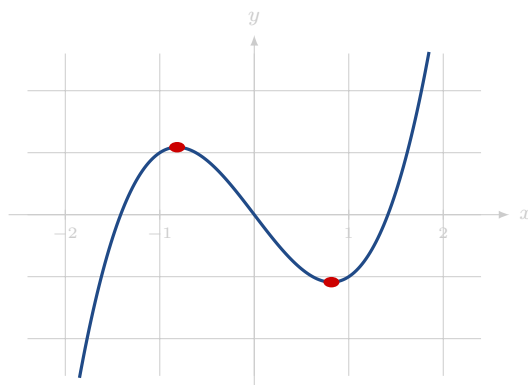
Recall: L1.4 The Derivative

Name _____

Solo – 9 minutes

1. $f(x) = x^2 - 3x$. Find $f'(x)$ **from the definition**.

2. The graph of f is below.



- (a) At the two marked points, what is f' ? (b) On what interval is f' negative? (c) Sketch f' on the same axes.

3. You found $\frac{d}{dx}(x^2) = 2x$ and $\frac{d}{dx}(x^3) = 3x^2$ from the definition. **Without doing any limits**, predict

$$\frac{d}{dx}(x^7) = \underline{\hspace{2cm}}$$

Then write down the rule you just used, for x^n .